Report I - SF2930 Regression Analysis

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Project 1

Scenario I: – Body mass fat data ($p < n$) - BFM men

Introduction and Project Goals

In this project, **Scenario I: Body mass fat data ($p < n$)** was selected, utilizing the **BFM men** dataset consisting of measurements from 252 men. The dataset includes body density as the response variable, along with a range of anthropometric measurements serving as potential predictors. The primary objective of the study is to develop a robust model, able to effectively and accurately predict body density using these explanatory variables, through linear regression techniques.

The analysis begins with the estimation of a full multiple linear regression model incorporating all available predictors, followed by outlier evaluation and multicollinearity assessment. Subset selection is then carried out to identify the best candidate models, which are then compared and evaluated to find the optimal regression model for this scenario.

All code was implemented in `R`.

1 Analyses and Model Development

```
Residuals:
      Min        1Q    Median        3Q       Max
-0.021527 -0.007717  0.000096  0.006491  0.034114

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.139e+00  4.030e-02  28.248  < 2e-16 ***
age         -1.203e-04  7.515e-05  -1.601  0.11062
weight       2.395e-04  1.243e-04   1.926  0.05528 .
height       1.498e-04  2.230e-04   0.672  0.50243
neck         1.075e-03  5.401e-04   1.991  0.04765 *
chest        1.232e-04  2.303e-04   0.535  0.59339
abdomen     -2.277e-03  2.008e-04 -11.335  < 2e-16 ***
hip          5.513e-04  3.390e-04   1.626  0.10521
thigh       -6.149e-04  3.354e-04  -1.833  0.06799 .
knee        -4.844e-05  5.622e-04  -0.086  0.93141
ankle       -6.314e-04  5.145e-04  -1.227  0.22094
biceps      -5.755e-04  3.976e-04  -1.448  0.14907
forearm     -1.017e-03  4.626e-04  -2.198  0.02891 *
wrist        3.959e-03  1.243e-03   3.185  0.00164 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01 on 238 degrees of freedom
Multiple R-squared:  0.7381,    Adjusted R-squared:  0.7238
F-statistic:  51.6 on 13 and 238 DF,  p-value: < 2.2e-16
```

Figure 1: R-Summary of full model.

We can see in Figure 1 that we have at least one significant regressor since the overall p-value is very small. We can see that we have four regressors that are significant on the 5% level or lower. These regressors are:

- **neck** [$\Pr(>|t|) = 0.04765$]
- **abdomen** [$\Pr(>|t|) < 2e-16$]
- **forearm** [$\Pr(>|t|) = 0.02891$]
- **wrist** [$\Pr(>|t|) = 0.00164$].

We can further note that **thigh** and **weight** are moderately significant since the significance of these are on the 10% level. The R^2 and adjusted R^2 are fairly high indicating that the model is pretty good with all coefficients involved. From the t-values for the respective coefficient we can see that the most important coefficient seems to be **abdomen** ($|t_{\text{abdomen}}| = 11.335$).

In Figure 2 we see the *Variance Inflation Factors (VIF)* for the coefficients.

```
     age    weight    height      neck     chest   abdomen       hip     thigh      knee     ankle    biceps   forearm     wrist
2.250450 33.509320  1.674591  4.324463  9.460877 11.767073 14.796520  7.777865  4.612147  1.907961  3.619744  2.192492  3.377515
```

Figure 2: Variance Inflation Factors for all coefficients.

According to Section 9.4.2 [MPV], a VIF exceeding 5 or 10 indicates that the corresponding coefficient is poorly estimated because of multicollinearity. We can thus see from Figure 2 that the coefficients exceeding a VIF of 10 are **weight**, **hip** and **abdomen**, indicating *serious* multicollinearity. We can also see that coefficients exceeding a VIF of 5 are **chest** and **thigh**, indicating *high* multicollinearity. Since the highest VIF is given by **weight** at a value of 33.509320, which is much larger than the others, **weight** is most likely correlated with the other mentioned coefficients with high VIF's.

In Figure 3 we see for the left subfigure (*Residuals vs Fitted*) that a horizontal band pattern is apparent indicating constant variance for the full model. The normal plot (*Q-Q Residuals*), in the middle subfigure of Figure 3, no severe deviation from the straight line for the tails, indicating a normal distribution for the residuals. For the right subfigure in Figure 3 we can see a slight upward trend across the fitted values but since the line is moderately horizontal, constant variance seems to be apparent.

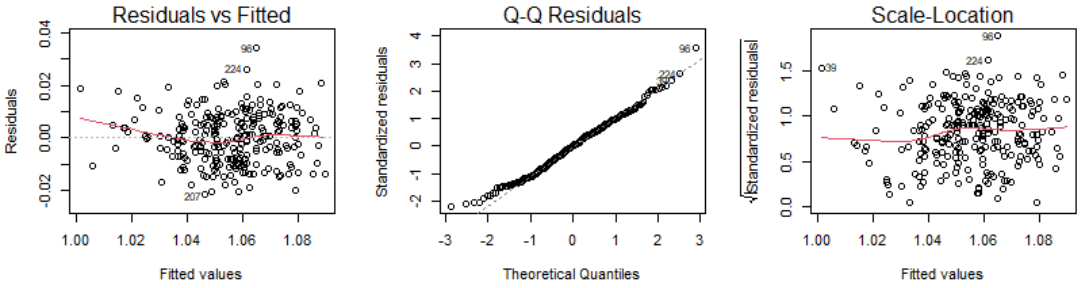
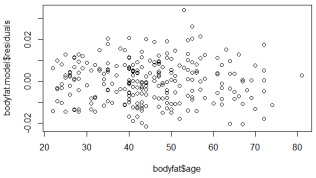
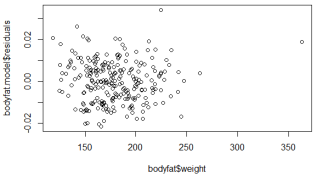


Figure 3: Residual plots for the full model.

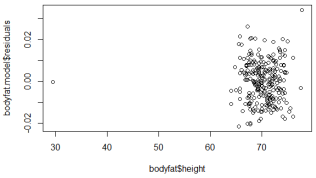
In Figure 4 we can see the residual plots for each of the regressor variables for the full model. The plots indicate that the variance is moderately constant since a horizontal band pattern seems to be apparent for each of the regressors, and thus, no transformation needs to be implemented.



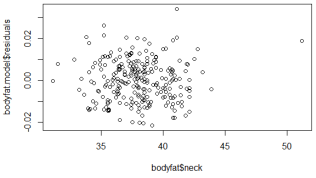
(a) Age



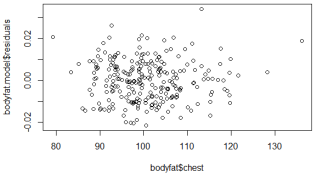
(b) Weight



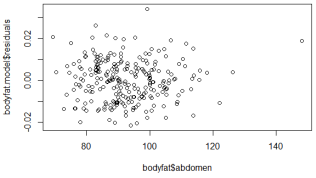
(c) Height



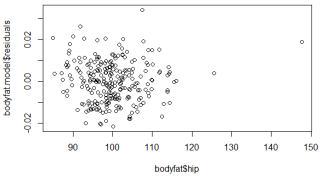
(d) Neck



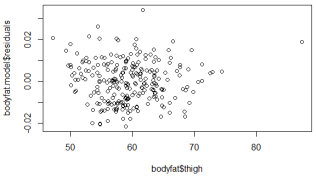
(e) Chest



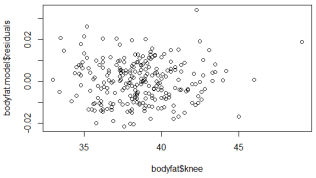
(f) Abdomen



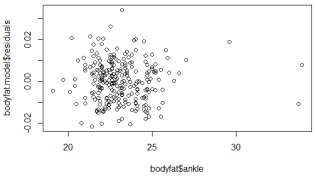
(g) Hip



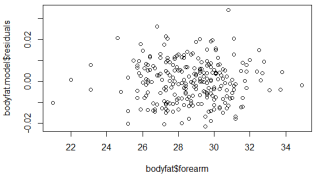
(h) Thigh



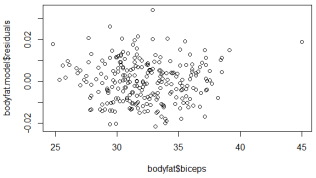
(i) Knee



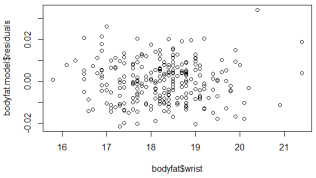
(j) Ankle



(k) Forearm



(l) Biceps



(m) Wrist

Figure 4: Residual plots for all explanatory variables.

For both the plots in Figure 3, and the plots in Figure 4, we can notice a few outlier points. The point 96 can clearly be observed as an outlier and is seen in the respective plots in Figure 3. The point 224 also seems to appear in all plots as an outlier, but is not as deviant as the point 96. The point 39 is also observed to be an outlier from the Scale-Location plot in Figure 3. We can further note that some outlier points can be seen in Figure 4, especially in plot (c) for the **height**, where the far left point (smallest **height**) corresponds to point 42.

To assess the influence of these points, we can look at *Cook's distance* and compare it to the *Cook's distance cutoff value* $F_{0.5,p,n-p} \approx 0.96$, according to Section 6.3 [MPV]. Cook's distance for all points, and the respective potential influential points, are marked with their index in Figure 5.

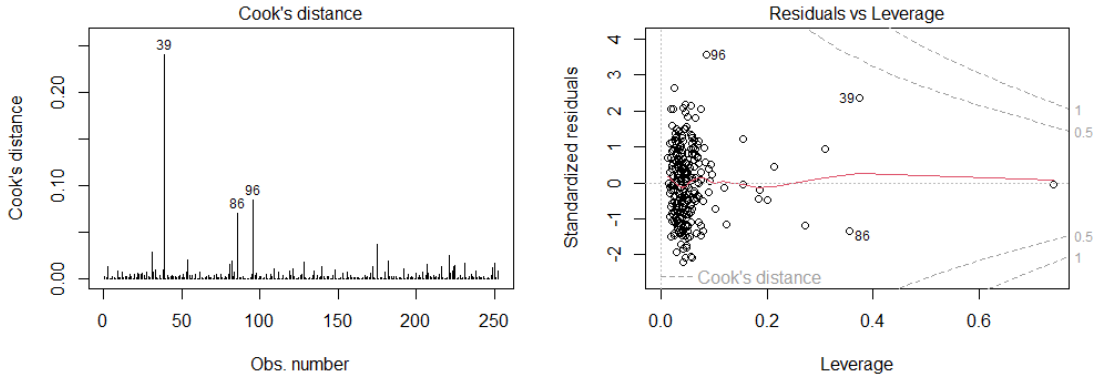


Figure 5: Cook's distance for all points with outliers marked with the respective data point number.

In both plots of Figure 5, we can see that the previously mentioned points, 39 and 96, are some of the points showing a greater Cook's distance than the other points (point 39 is further identified as the outlier we see in Figure 4 plot (b) for **weight** and point 96 is one outlier in Figure 4 plot (c) for the largest **height**). We can further see that the point 86 has a greater Cook's distance than the other points as well. Although these points are more influential than the others, none of the points have a Cook's distance exceeding the cutoff value, indicating that they are not very influential. We can also see that the points 42 and 224 seem to have very small influence on the fitted model.

We can also look at the $DFBETAS$ and $DFFITS$ to observe the effect of the data points on the estimated coefficients and fitted values respectively. For $DFBETAS$, we use a cutoff value of $2/\sqrt{n} \approx 0.13$ and for $DFFITS$ we use a cutoff value of $2\sqrt{p/n} \approx 0.47$ as recommended in Section 6.4 [MPV]. The $DFBETAS_{i,j}$ values, for data point i and regressor j, can be seen for some potential outliers in Table 1, where the values that are marked red indicate that $|DFBETAS_{i,j}| > 0.13$. The $DFFITS_i$ values, for data point i, can also be seen in Table 2 for the same potential outliers, where the values that are

marked red indicate that $|DFFITS_i| > 0.47$.

To see if the points are influential in a precision-of-estimation sense, we can look at the $COVRATIO$ values for the potential outliers. For $COVRATIO$, we used a lower bound cutoff value of $1-3p/n \approx 0.83$ and an upper bound cutoff value of $1+3p/n \approx 1.17$. The $COVRATIO_i$ values can be seen in Table 2 where the red values indicate that $0.833 < |COVRATIO_i|$ or $|COVRATIO_i| > 1.17$.

Table 1: DFBETAS values for potential outliers and for all regressor variables. Red values have an absolute value larger than the cutoff value 0.13.

	int.	age	weight	height	neck	chest	abdomen	hip	thigh	knee	ankle	biceps	forearm	wrist
5	-0.01	0.02	0.00	0.00	0.02	0.01	-0.03	0.02	0.00	-0.02	0.00	-0.01	0.00	-0.01
31	-0.14	-0.03	-0.13	0.07	0.12	0.05	0.04	0.07	-0.09	-0.07	0.61	0.06	-0.07	-0.13
36	0.11	0.01	0.13	-0.05	-0.03	-0.11	0.00	-0.16	0.05	0.01	-0.01	0.02	-0.05	0.06
39	0.36	0.26	0.66	-0.37	0.42	-0.58	-0.26	0.34	-0.23	-0.37	0.11	0.08	-0.75	-0.26
41	-0.05	-0.01	0.03	0.00	-0.04	0.04	0.01	0.09	0.02	-0.14	0.03	-0.06	0.03	0.10
42	-0.05	0.00	-0.03	0.09	0.01	0.02	0.02	0.01	0.01	-0.01	0.01	0.01	0.00	0.00
86	0.07	-0.26	0.08	0.02	0.00	0.01	-0.02	-0.04	0.06	0.17	-0.95	-0.14	0.00	0.24
96	-0.11	0.14	0.26	0.15	-0.13	0.34	-0.53	0.07	-0.08	0.13	-0.49	-0.44	0.04	0.35
106	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-0.01
159	0.00	0.01	0.00	0.01	-0.01	0.01	-0.01	-0.01	0.02	0.01	0.00	0.03	-0.08	0.02
175	0.12	0.21	0.13	-0.03	-0.08	-0.19	0.05	-0.06	-0.05	-0.08	0.10	-0.12	0.65	-0.29
206	-0.04	0.02	-0.05	0.02	-0.04	0.01	0.02	0.06	-0.01	0.00	0.01	-0.08	0.18	-0.03
224	0.07	0.24	0.01	0.02	-0.04	0.01	-0.05	0.00	0.13	-0.18	0.18	0.00	0.03	-0.16

Table 2: $DFFITS$ and $COVRATIO$ values for potential outliers. Red values have an absolute value larger than the cutoff value 0.47.

	$DFFITS$	$COVRATIO$
5	-0.05	1.20
31	0.63	1.46
36	-0.23	1.31
39	1.85	1.21
41	0.23	1.33
42	-0.10	4.08
86	-0.99	1.48
96	1.11	0.54
106	-0.02	1.26
159	-0.09	1.30
175	-0.72	1.34
206	-0.21	1.28
224	0.43	0.72

From Table 1 and Table 2 we see that the previously mentioned points $39, 42, 86$ and 96 seem to be outliers which are rather influential, and we therefore continue to evaluate how the removal of these points affect the model.

First we remove data point number 42 from the data set since it seems to be a faulty measurement with a height of 30 inches or 72.6 cm. We came to this conclusion since the other measurements for this data point are close to average for the set. Thus for the rest of this project data point number 42 is excluded (we realized later that this in fact was a faulty measurement from the data description and that one should change the **height** of point 42 from 29.5 inches to 69.5 inches; nevertheless, the model is not affected that much by the removal of this point and the conclusion regarding the removal of the other points holds).

To observe what happens to the model for the removal of these points, we look at the reduction in MS_{Res} for the removal of all cases with the three points $39, 86$ and 96. Table 3 gives the values for the decrease in % in MS_{Res} given by

$$\frac{MS_{Res} - MS_{Res_{(i)}}}{MS_{Res}} \times 100$$

where $MS_{Res_{(i)}}$ gives the residual mean square after removing observations indexed by

$$i \in \mathcal{P}\left(\{39, 86, 96\}\right) \setminus \{\varnothing\}$$

where $\mathcal{P}$ is the power set.

Table 3: Decrease in model MS_{Res}, given in %, after the removal of combinations of the selected influential points.

(86)	-0.09%
(39)	1.85%
(39, 86)	2.02%
(96)	4.59%
(86, 96)	5.38%
(39, 96)	6.87%
(39, 86, 96)	7.45%

We can from Table 3 see that removing all of these points gives the largest decrease in MS_{Res} and therefore we remove all these observations, considering also what has been previously stated.

After removing the outliers, the focus shifts to assessing whether any of the regressors require transformation. This is done to ensure that the key assumptions of Linearity and homoscedasticity hold for the linear regression as per Chapter 5 [MPV].

From Figure 6 and Figure 7 it can be concluded that no transformations of the regressors are necessary. The residual plots do not exhibit any systematic patterns or signs of non-constant variance, indicating that the assumptions of linearity and homoscedasticity are reasonably satisfied.

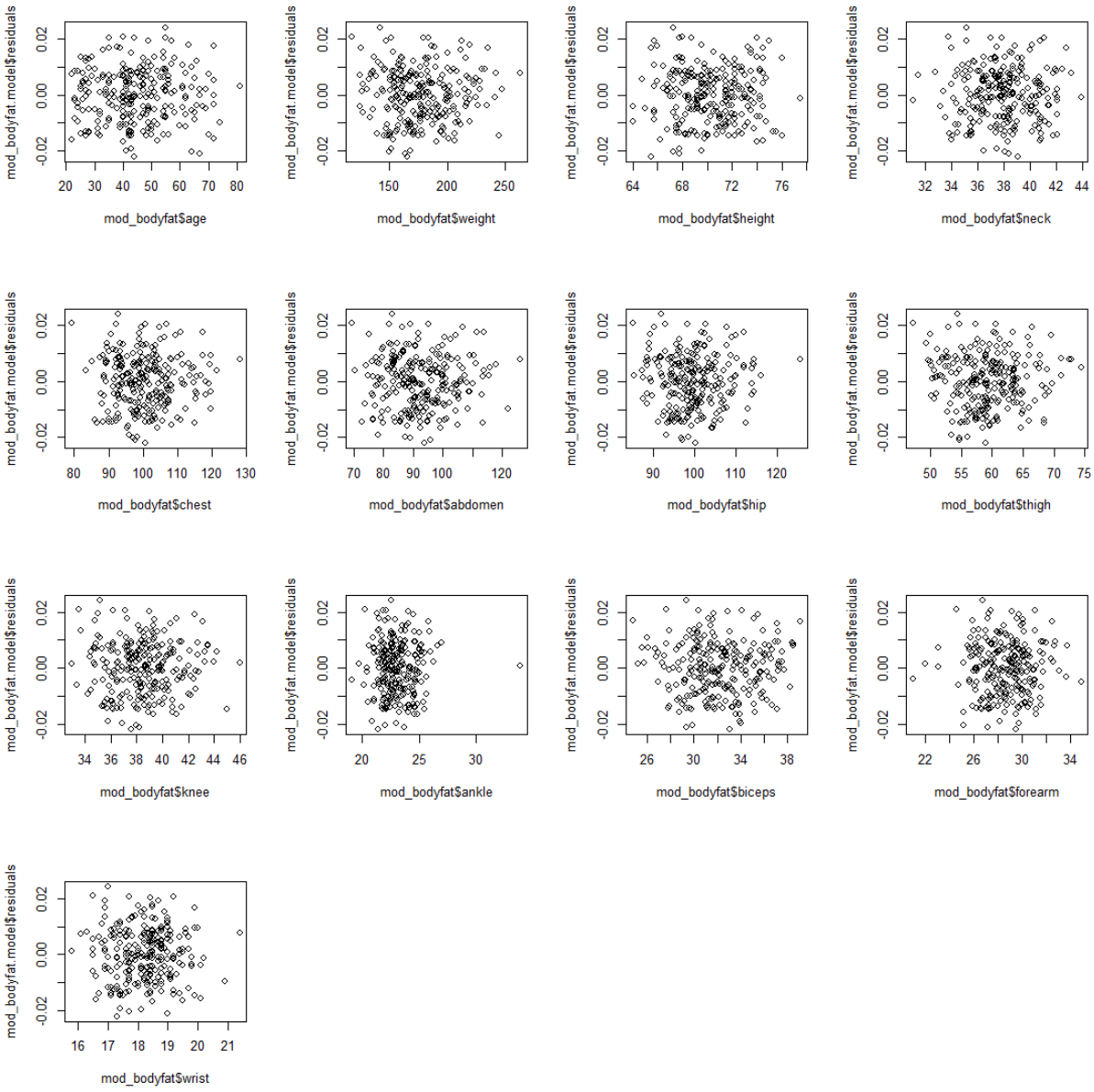


Figure 6: Residual plots after removing outliers.

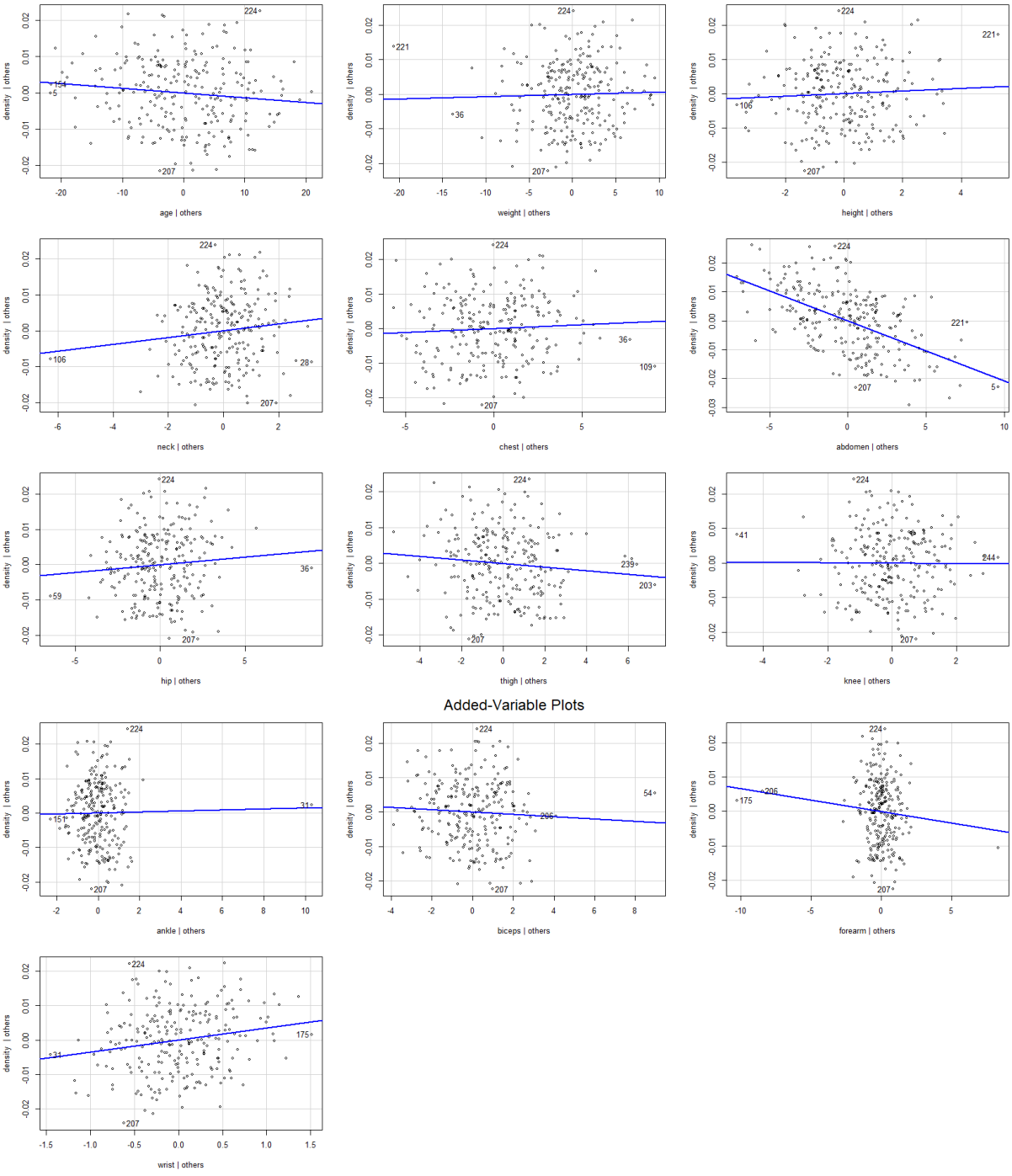


Figure 7: Added variable plots showing the partial effect of each regressor after accounting for all other regressors.

In the next step of model development, the goal is to find the best subset model by ranking them by their respective Mallow's C_p adjusted R^2, and the PRESS statistics.

The `regsubsets` function from the `leaps` package was applied to evaluate all possible subsets of regressors, identifying the best model for each subset size based on Residual Sum of Squares (RSS).

From Table 4, models of interest were identified as the models with 6, 7 and 8 regressors.

Table 4: Model comparison across all feasible regressions.

# of regressors	R^2	Adjusted R^2	C_p
1	0.6843643	0.6830812	51.000396
2	0.7241721	0.7219204	15.795154
3	0.7348426	0.7315825	7.822210
4	0.7383254	0.7340180	6.567101
5	0.7406028	0.7352434	6.438578
6	0.7436248	0.7372421	5.614125
7	0.7458920	0.7384805	5.495169
8	0.7474665	0.7390135	6.023625
9	0.7485538	0.7390453	7.007433
10	0.7493020	0.7387240	8.308139
11	0.7495674	0.7378947	10.060064
12	0.7496160	0.7368305	12.014608
13	0.7496317	0.7357223	14.000000

For these indicated models, the PRESS values were calculated and are shown in Table 5.

Table 5: PRESS comparison for selected regression models.

Model	PRESS
Best 6-reg	0.02335639
Best 7-reg	0.02333158
Best 8-reg	0.02335534
All 13 reg	0.02406489

From Table 4 plotted in Figure 8 and Table 5, the model with 7 regressors emerged as the best, containing the following predictors: **age**, **weight**, **neck**, **abdomen**, **thigh**, **forearm**, **wrist**.

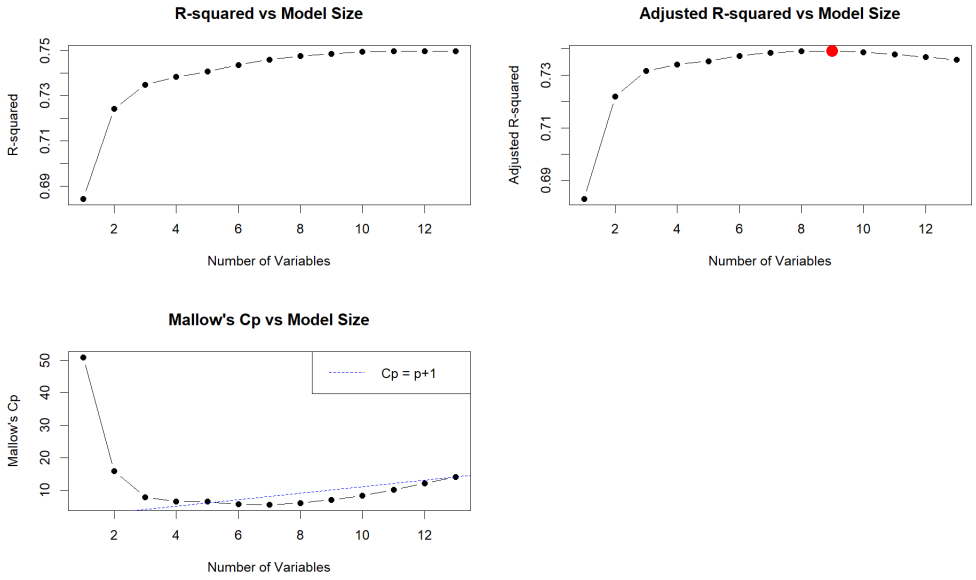


Figure 8: R^2, adjusted R^2 and Mallow's C_p.

From Table 5 the 7 regressor model was chosen for further investigation. To assess whether removing 6 regressors reduced the multicollinearity problem, the VIF scores of Model 7 were compared to those of the full model. A VIF score exceeding 10 indicates problematic multicollinearity, meaning the coefficient estimates are unreliable due to high correlation between regressors.

Table 6: Comparison of Variance Inflation Factors (VIF) for the full and selected regression models.

Regressor	Full model	Best model (7 regressors)
age	2.337601	2.089586
weight	44.744845	12.549217
height	3.355012	-
neck	3.944037	3.555888
chest	10.754262	-
abdomen	12.094542	7.084485
hip	12.079516	-
thigh	7.146980	4.546464
knee	4.457598	-
ankle	2.316505	-
biceps	3.488859	-
forearm	2.411169	2.148051
wrist	3.347682	2.955719

From Table 6 one can first observe that the seven predictor model is far superior in terms of VIF from the original full model. Furthermore a VIF greater than 10 is observed for the predictor **weight** in the subsequent modified model. Thus, further investigation is done to examine if it can be removed from the model, according to standard practice.

In this 7 regressor model, weight has by far the largest VIF score (12.549217), indicating severe multicollinearity with the other retained regressors, likely because body weight is strongly correlated with measures like **abdomen**, **thigh**, and **neck** circumference that are already in the model. Thus removing the **weight** regressor was investigated.

Table 7: VIF for model without problematic weight regressor.

Age	Neck	Abdomen	Thigh	Forearm	Wrist
1.827932	3.375709	3.922031	3.930678	2.080710	2.466586

With the information in Table 7 in mind no further action is justified in regards to multicollinearity concerns.

Since the function `regsubsets` usually tends to favor larger models, it was further investigated how further reduced models performed. In Figure 9 we observe that for the 6-regressor model, thigh and forearm are not significant. Removing **thigh** and **forearm** results in a model in which **neck** is not significant, and removing **neck** results in a very simple model with only **age**, **abdomen**, and **wrist** as its variables. Notably, this model seems to have almost the same predictive power as the 6-regressor model.

	Dependent variable:		
	density		
	(age + neck + abdomen + thigh + forearm + wrist) (1)	(age + neck + abdomen + wrist) (2)	(age + abdomen + wrist) (3)
age	-0.0002***	-0.0001***	-0.0002***
	(0.0001)	(0.0001)	(0.0001)
neck	0.001**	0.001	
	(0.0005)	(0.0005)	
abdomen	-0.002***	-0.002***	-0.002***
	(0.0001)	(0.0001)	(0.0001)
thigh	-0.0003		
	(0.0002)		
forearm	-0.001		
	(0.0004)		
wrist	0.005***	0.004***	0.005***
	(0.001)	(0.001)	(0.001)
Constant	1.129***	1.124***	1.130***
	(0.014)	(0.013)	(0.013)
Observations	248	248	248
R2	0.738	0.734	0.731
Adjusted R2	0.731	0.729	0.728
Residual Std. Error	0.010 (df = 241)	0.010 (df = 243)	0.010 (df = 244)
F Statistic	112.947*** (df = 6; 241)	167.312*** (df = 4; 243)	220.836*** (df = 3; 244)
Note:			*p<0.1; **p<0.05; ***p<0.01

Figure 9: Comparison of reduced models.

In Table 8, we can see that that the differences between the $PRESS$ value for the models are negligible, indicating that the models, including the original model, have similar predictive power. We can see from the values for C_p, where the reference was the full model (which is not necessarily the true model), that there is little difference between the reduced models, also suggesting that the bias-variance tradeoff is similar for the models. We can further see that the AIC have negligible difference between the models and that BIC favors the simplest model, further suggesting that there is not a big loss in predictive ability between the simplest model and the others.

Table 8: $PRESS$, C_p, AIC and BIC comparison between original model and subset models.

Model	PRESS	C_p	AIC	BIC
Original Model	0.0241	14.0	-1583.9	-1531.2
(**age** + **neck** + **abdomen** + **thigh** + **forearm** + **wrist**)	0.0239	11.2	-1586.3	-1558.2
(**age** + **neck** + **abdomen** + **wrist**)	0.0240	11.0	-1586.5	-1565.4
(**age** + **abdomen** + **wrist**)	0.0240	11.6	-1585.9	-1568.4

Since **neck** was considered insignificant, and the models seem to have about the same predictive ability, by Occam's razor the simple 3-regressor model is considered the *best* regressor model.

Results

In the first step of this project, a full model with all data points and all predictors was generated. From this, some candidates for significant predictors and troublesome outlier points were identified. Firstly, the data point 42 was removed due to likelihood it was a faulty point through analysis. With the information from Table 3 it was concluded that removing the combination of candidate points $\{39, 86, 96\}$ resulted in the largest decrease in MS_{Res} for the model, and was therefore removed from the set of data points. This first change in the model gave the following results

```
Residuals:
      Min        1Q    Median        3Q       Max
-0.021527 -0.007717  0.000096  0.006491  0.034114

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.139e+00  4.030e-02  28.248  < 2e-16 ***
age         -1.203e-04  7.515e-05  -1.601  0.11062
weight       2.395e-04  1.243e-04   1.926  0.05528 .
height       1.498e-04  2.230e-04   0.672  0.50243
neck         1.075e-03  5.401e-04   1.991  0.04765 *
chest        1.232e-04  2.303e-04   0.535  0.59339
abdomen     -2.277e-03  2.008e-04 -11.335  < 2e-16 ***
hip          5.513e-04  3.390e-04   1.626  0.10521
thigh       -6.149e-04  3.354e-04  -1.833  0.06799 .
knee        -4.844e-05  5.622e-04  -0.086  0.93141
ankle       -6.314e-04  5.145e-04  -1.227  0.22094
biceps      -5.755e-04  3.976e-04  -1.448  0.14907
forearm     -1.017e-03  4.626e-04  -2.198  0.02891 *
wrist        3.959e-03  1.243e-03   3.185  0.00164 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.01 on 238 degrees of freedom
Multiple R-squared:  0.7381,    Adjusted R-squared:  0.7238
F-statistic:  51.6 on 13 and 238 DF,  p-value: < 2.2e-16
```

Figure 10: Full model.

```
Residuals:
      Min        1Q    Median        3Q       Max
-0.022025 -0.006967  0.000218  0.006657  0.024146

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.108e+00  5.299e-02  20.907  < 2e-16 ***
age         -1.289e-04  7.421e-05  -1.737  0.08370 .
weight       7.029e-05  1.518e-04   0.463  0.64382
height       3.748e-04  4.350e-04   0.862  0.38983
neck         9.481e-04  5.312e-04   1.785  0.07557 .
chest        2.189e-04  2.464e-04   0.888  0.37543
abdomen     -2.077e-03  2.080e-04  -9.985  < 2e-16 ***
hip          4.313e-04  3.301e-04   1.307  0.19262
thigh       -4.954e-04  3.326e-04  -1.490  0.13769
knee        -6.762e-05  5.594e-04  -0.121  0.90390
ankle        1.444e-04  6.162e-04   0.234  0.81487
biceps      -3.441e-04  3.898e-04  -0.883  0.37824
forearm     -6.702e-04  4.680e-04  -1.432  0.15345
wrist        3.533e-03  1.235e-03   2.861  0.00461 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.009622 on 234 degrees of freedom
Multiple R-squared:  0.7496,    Adjusted R-squared:  0.7357
F-statistic: 53.89 on 13 and 234 DF,  p-value: < 2.2e-16
```

Figure 11: Model with removed points.

In Figure 10 and Figure 11, an improved R^2 and adjusted R^2 is observed. From earlier in Table 3 the MS_{Res} value also decreased by 7.45%. Otherwise, the model is deemed to be similar, which implies that the removal of the points has a positive, or at worst neutral, impact on the model.

In Figure 12 and Figure 13 below, we observe the difference between the full model and the model with removed outliers. We can see that removing the influential points effectively results in the plots looking closer to those of the normally distributed case.

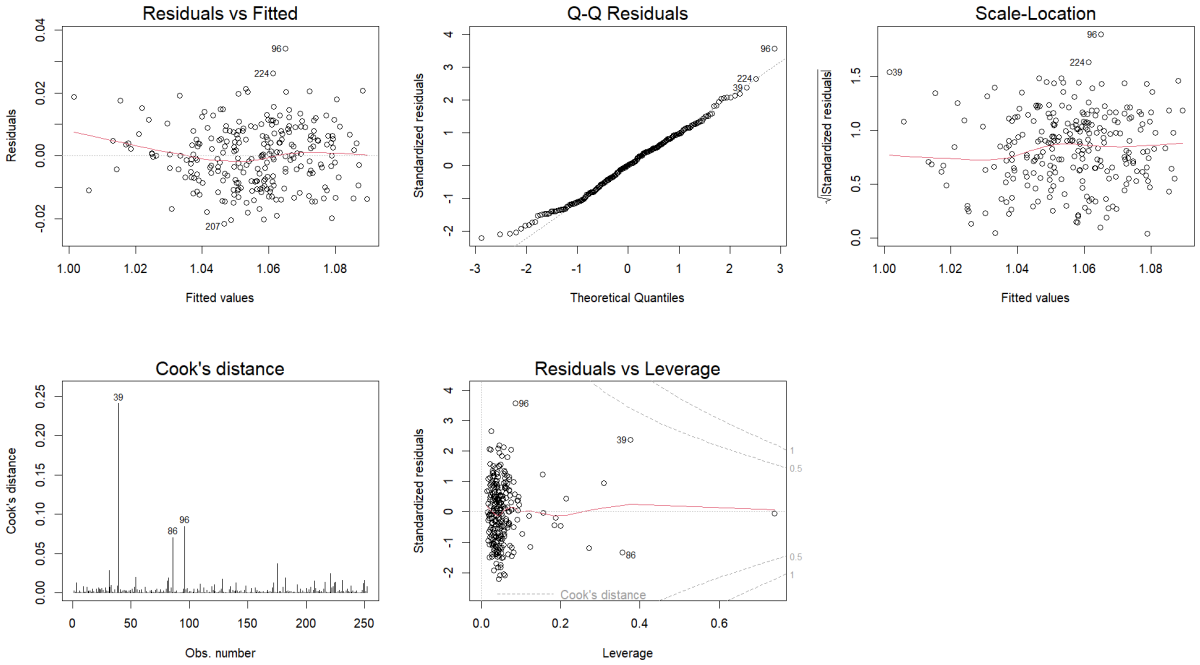


Figure 12: Original model.

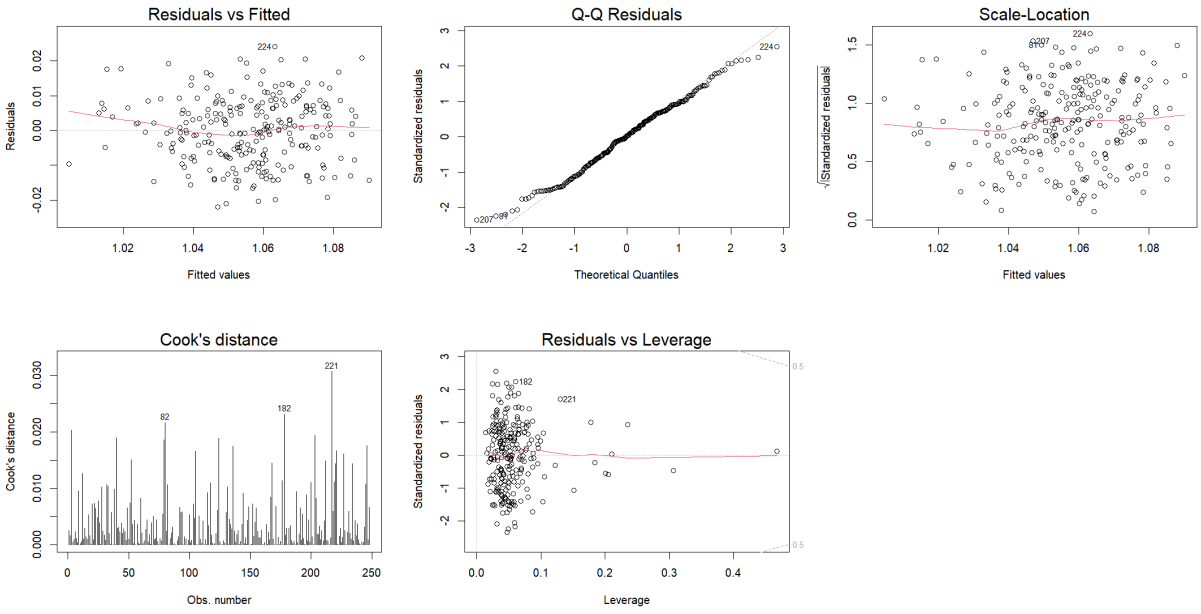


Figure 13: Model after removal of outliers.

As mentioned before, according to analysis of Figure 6 it was concluded that no transformation of the response or any of the regressors was necessary.

From Figure 9 and Table 8 it was evident that the simplest model, with the regressor variables **age**, **abdomen** and **wrist**, was considered the *best* model since its predictive power is almost the same as the other models while it is also a very simple model. The ANOVA summary for the original model and this final model can be seen in Figure 14 and Figure 15 below. Comparing Figure 14 with Figure 15 we can see that we have a much simpler model with only significant regressor coefficients included, and with approximately the same predictive power. Worth to note is also that all regressor coefficients have very low VIF values in the range from 1 to 2 indicating no multicollinearity issue. The difference in adjusted R^2 is only a decrease of 1.1% and the residual standard error increase is only 1.5%.

```
Residuals:
      Min        1Q    Median        3Q       Max
-0.022025 -0.006967  0.000218  0.006657  0.024146

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.108e+00  5.299e-02  20.907  < 2e-16 ***
age         -1.289e-04  7.421e-05  -1.737  0.08370 .
weight       7.029e-05  1.518e-04   0.463  0.64382
height       3.748e-04  4.350e-04   0.862  0.38983
neck         9.481e-04  5.312e-04   1.785  0.07557 .
chest        2.189e-04  2.464e-04   0.888  0.37543
abdomen     -2.077e-03  2.080e-04  -9.985  < 2e-16 ***
hip          4.313e-04  3.301e-04   1.307  0.19262
thigh       -4.954e-04  3.326e-04  -1.490  0.13769
knee        -6.762e-05  5.594e-04  -0.121  0.90390
ankle        1.444e-04  6.162e-04   0.234  0.81487
biceps      -3.441e-04  3.898e-04  -0.883  0.37824
forearm     -6.702e-04  4.680e-04  -1.432  0.15345
wrist        3.533e-03  1.235e-03   2.861  0.00461 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.009622 on 234 degrees of freedom
Multiple R-squared:  0.7496,    Adjusted R-squared:  0.7357
F-statistic: 53.89 on 13 and 234 DF,  p-value: < 2.2e-16
```

Figure 14: Original model with all regressors.

```
Residuals:
       Min         1Q     Median         3Q        Max
-0.0233364 -0.0077002  0.0008551  0.0068237  0.0230678

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.130e+00  1.273e-02  88.755  < 2e-16 ***
age         -1.583e-04  5.103e-05  -3.102  0.00215 **
abdomen     -1.739e-03  7.699e-05 -22.591  < 2e-16 ***
wrist        5.134e-03  8.627e-04   5.951  9.2e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.00977 on 244 degrees of freedom
Multiple R-squared:  0.7308,    Adjusted R-squared:  0.7275
F-statistic: 220.8 on 3 and 244 DF,  p-value: < 2.2e-16
```

Figure 15: Final reduced model.

To estimate the precision of the model estimates, the *bootstrap method* was used. In Figure 16, all bootstrap distributions appear approximately normal and symmetric, which validates the use of standard confidence intervals and suggests the estimates are stable.

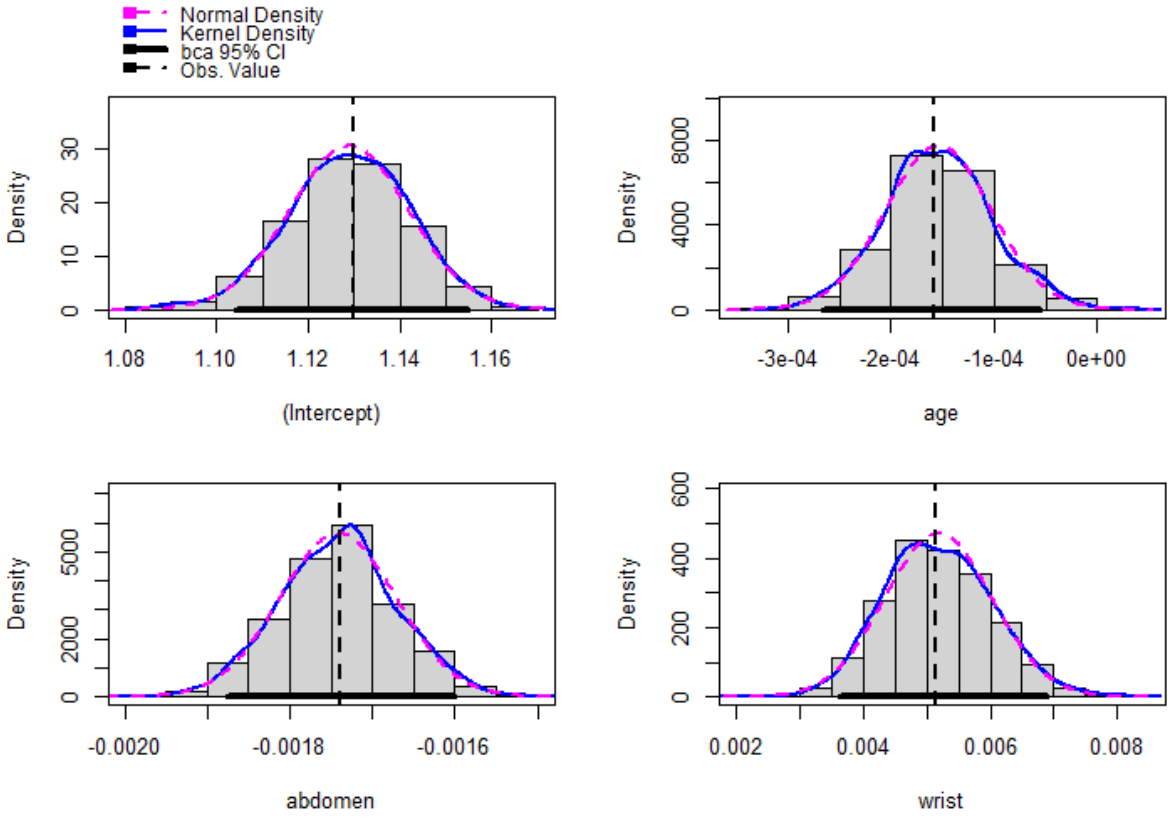


Figure 16: Bootstrap estimate distributions for final model.

In addition, the bootstrap confidence intervals for each regressor coefficient in the final model can be seen in Figure 17, corresponding to the black solid line in Figure 16.

	2.5 %	97.5 %
(Intercept)	1.1041730890	1.154843e+00
age	-0.0002662343	-5.673156e-05
abdomen	-0.0018753530	-1.600361e-03
wrist	0.0036301842	6.886506e-03

Figure 17: Bootstrap confidence interval for final model

The intervals confirm that all the regressor coefficients for **age**, **abdomen** and **wrist** are statistically significant since none of the confidence intervals include zero, and which also falls in line with what can been seen in Figure 15.

In Figure 18 we can also see bootstrapped residual plots for the final model, where we can see that the model behaves adequately.

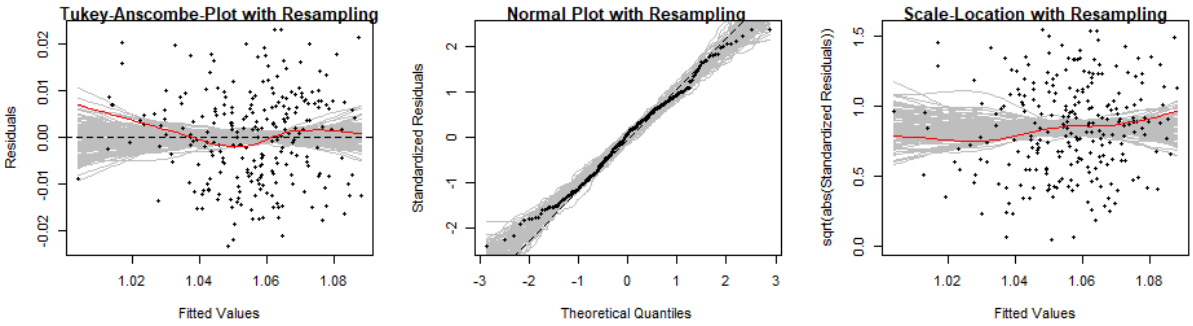


Figure 18: Bootstrapped residual plots of the final model (using `resplots.R`)

Conclusion

The final model is a three-predictor model, including **age**, **abdomen** and **wrist**. The findings of this analysis indicate that these variables are the most statistically significant predictors of a man's body density, as confirmed by bootstrap confidence intervals and various criterion. Although arguments can be made for models with a greater number of predictors, the selected model represents the most adequate and well-supported choice based on the criteria evaluated. This conclusion was reached following a thorough analysis of all possible subset models and their respective diagnostic measures. As observed in Figure 16 and Figure 17 all four distributions appear approximately normal and symmetric and all three regressors in the final model have confidence intervals that do not contain zero, confirming that all predictors are statistically significant at the 5% level. This shows the final model is robust. It is also clear in Figure 9 and **??** that the 3 predictor model, despite its simplicity, in comparison to the original model maintains a similar R^2 and adjusted R^2, explaining approximately 73% of the variance in body density, with a highly significant F-statistic of 220.8 and a residual standard error of 0.00977.